

Nilin Abrahamsen

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[GitHub](#) | [Twitter](#) | [Google Scholar](#)

Experience

Voleon Capital Management – Member of Research Staff, 2024–Present

- Led development and launch of a new portfolio strategy.
- Mentored intern to create alpha signals.

University of California, Berkeley – Postdoctoral Researcher, 2021–2024

- Developed optimization methods and neural architectures for computational physics.
 - Extensive use of ML frameworks (JAX, PyTorch) and JIT-compiled Python (Numba-CUDA).
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Education

Massachusetts Institute of Technology – PhD in Mathematics

- Tensor network algorithms for computational physics (Advisors: Peter Shor and Jon Kelner).

University of Copenhagen – BS and MS in Mathematics

- Awarded Best MS Thesis in Mathematics in Denmark (Danish Mathematical Association).
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Projects

NanoGPT speedrun, optimization track – May 2026

- Broke the overall record 3 times for the optimization track of the [NanoGPT speedrun](#).
- Introduced 2 innovations (contra-muon and [radial brake](#)) which withstood [10,000 ablation runs](#) by PrimeIntellect's autonomous agents.

PROMA: Projected Microbatch Accumulation for Reference-Free Proximal Policy Updates – Jan 2026 ([github](#), [arxiv:2601.10498](#))

- Proximal policy optimization without a reference policy.
- Projects away high-variance subspace of the policy gradient to approximate the natural gradient.
- Implemented in VeRL.

Inventing Art Styles without Human Training Data – May 2023 ([github](#), [arxiv:2305.12015](#))

- Demonstrated generative art without human-made training data (“art-zero”).
 - The artist model controls a paint brush, and artistic expression arises from representing input images under the constraints of the painting tool. Use of a reconstruction loss allows for abstraction. Trained with differentiable digital paintbrushes implemented in JAX using a straight-through estimator.
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Skills & Languages

Programming & Tools: PyTorch, JAX, Numba-CUDA, Reinforcement Learning, Context Engineering

Languages: English, Danish, Mandarin Chinese (professional working proficiency)